Akoizo Original Practice - Form 5

Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) What does a horizontal stabilizer mainly provide?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

A. Pitch stability

B. Thrust

C. Roll control

D. Lift for the whole plane

Answer: _____

2. (2 points) Why do many teams wind to a target torque?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

A. Torque predicts the climb rate better than turns alone

B. It is required

C. It breaks fewer winders

D. It adds energy

Answer: _____

3. (2 points) Which propeller is NOT allowed?

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

A. One whose blade angle changes by an active mechanism during flight

B. A purchased fixed-pitch propeller

C. A team-built balsa propeller

D. A propeller with a freewheel

Answer: _____

4. (2 points) Which practice keeps trim consistent between flights?

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

A. Changing propellers often

B. Retensioning film between flights

C. Marking and keying removable wing and tail positions

D. Storing the plane loose in a bag

Answer: _____

5. (4 points) Explain why blade angle changes from hub to tip.

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

Answer: _____
_____
_____

6. (4 points) Why do indoor planes use long, narrow wings?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

Answer: _____
_____
_____

7. (4 points) Why record bad flights in the log?

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

Answer: _____
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_____

8. (4 points) Why make only one trim change at a time?

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

Answer: _____
_____
_____

9. (4 points) What should the log record for each motor?

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

Answer: _____
_____
_____

10. (6 points) Solve this case and explain your reasoning: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?

Answer: _____
_____
_____

12. (2 points) Why did Prop 2 give a short flight?

Supplied synthetic observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

A. It needed more torque than the cruise portion supplied

B. It spun too fast

C. It was too light

Answer: _____

13. (2 points) Define “Lift equation” in the context of Wright Stuff.

Answer: _____
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_____

14. (2 points) Define “Controlled variable” in the context of Wright Stuff.

Answer: _____
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15. (2 points) Define “Neutral point” in the context of Wright Stuff.

Answer: _____
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16. (2 points) Define “Torque burst” in the context of Wright Stuff.

Answer: _____
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_____

17. (2 points) Define “Official flight” in the context of Wright Stuff.

Answer: _____
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18. (2 points) Define “Ballast” in the context of Wright Stuff.

Answer: _____
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